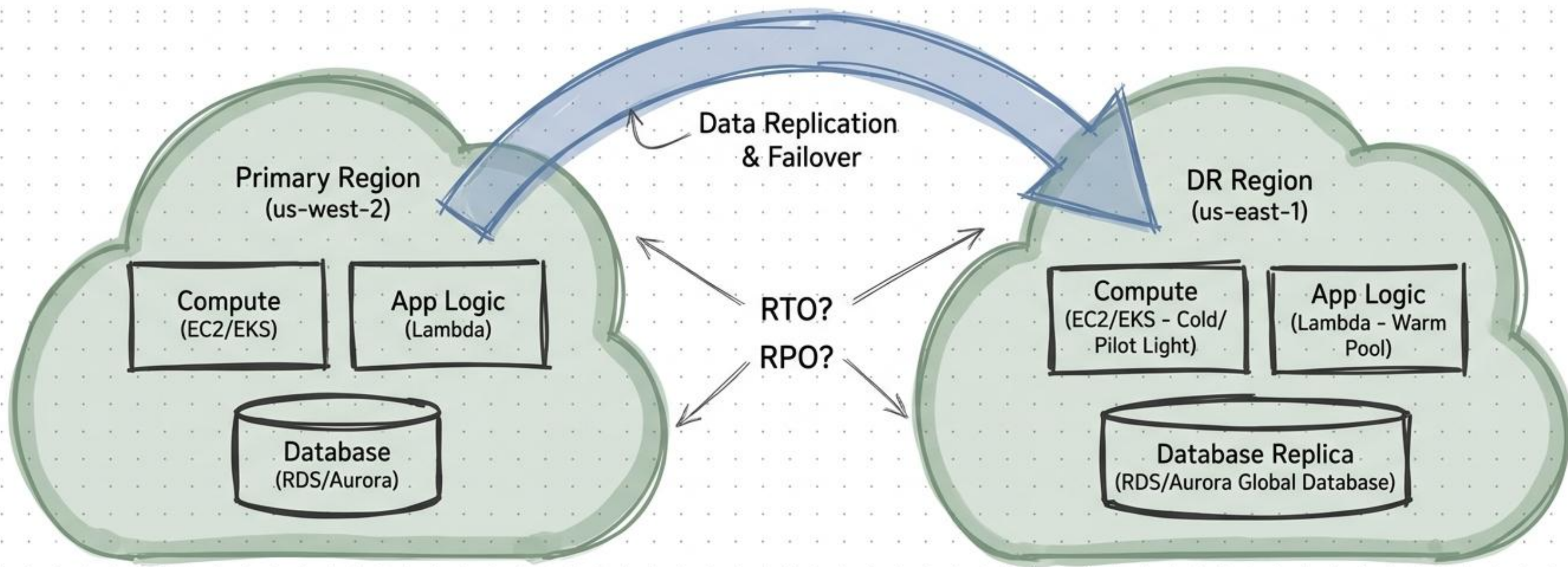


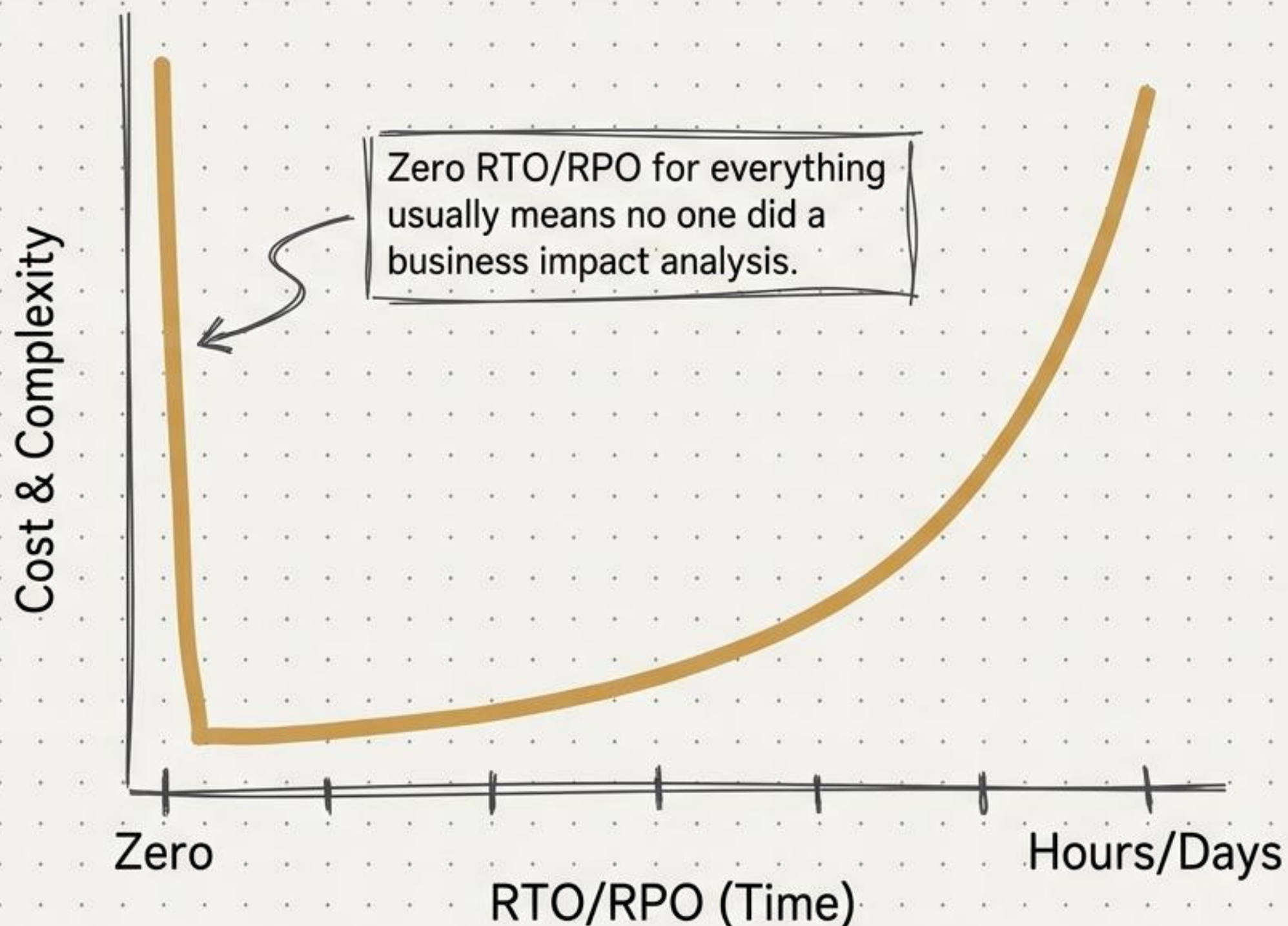
Designing AWS Multi-Region Disaster Recovery

A deterministic framework to route any application to the right DR strategy.

For Application Teams, Cloud Architects, and Business Owners.

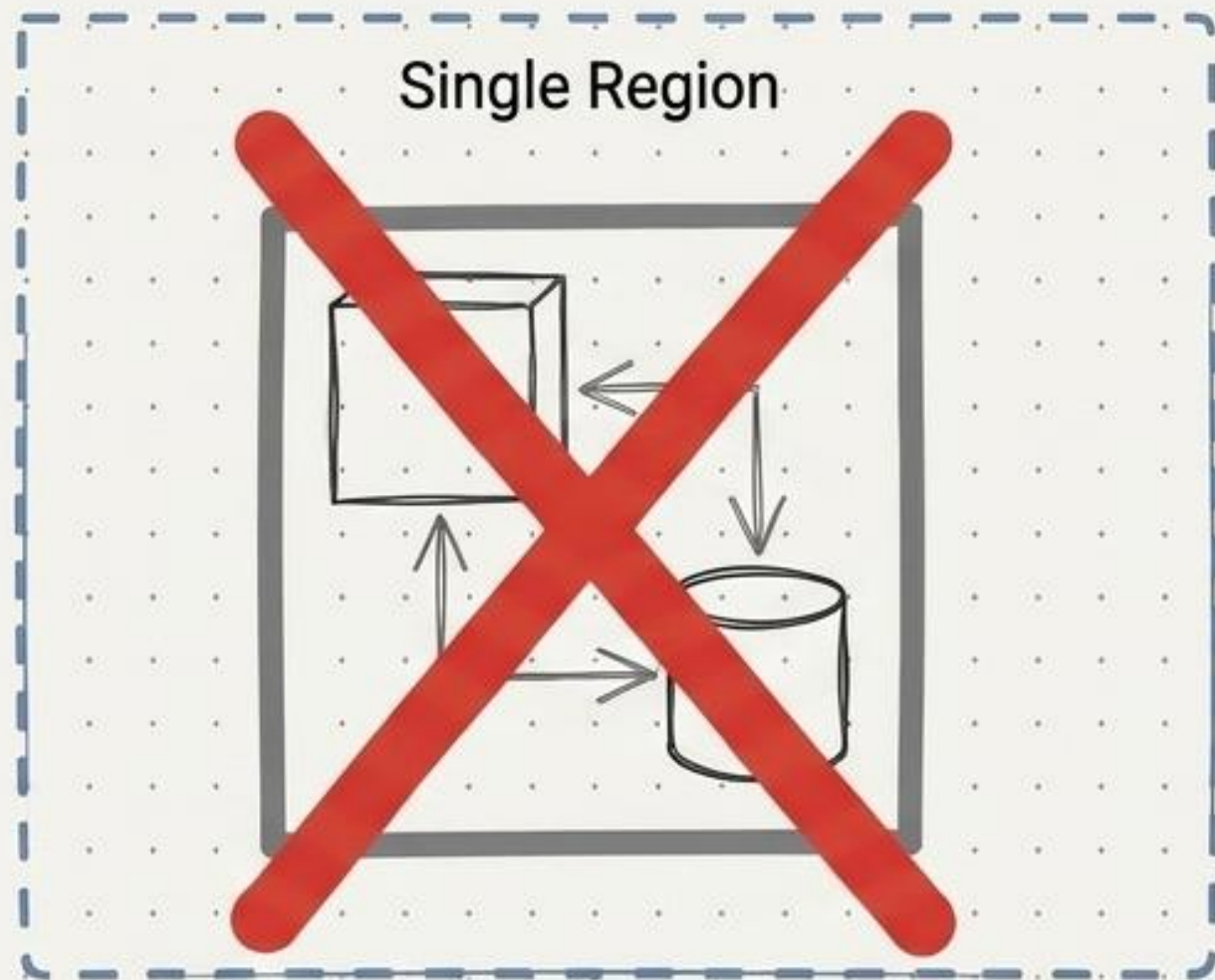


Pick the simplest strategy that meets your measured objectives

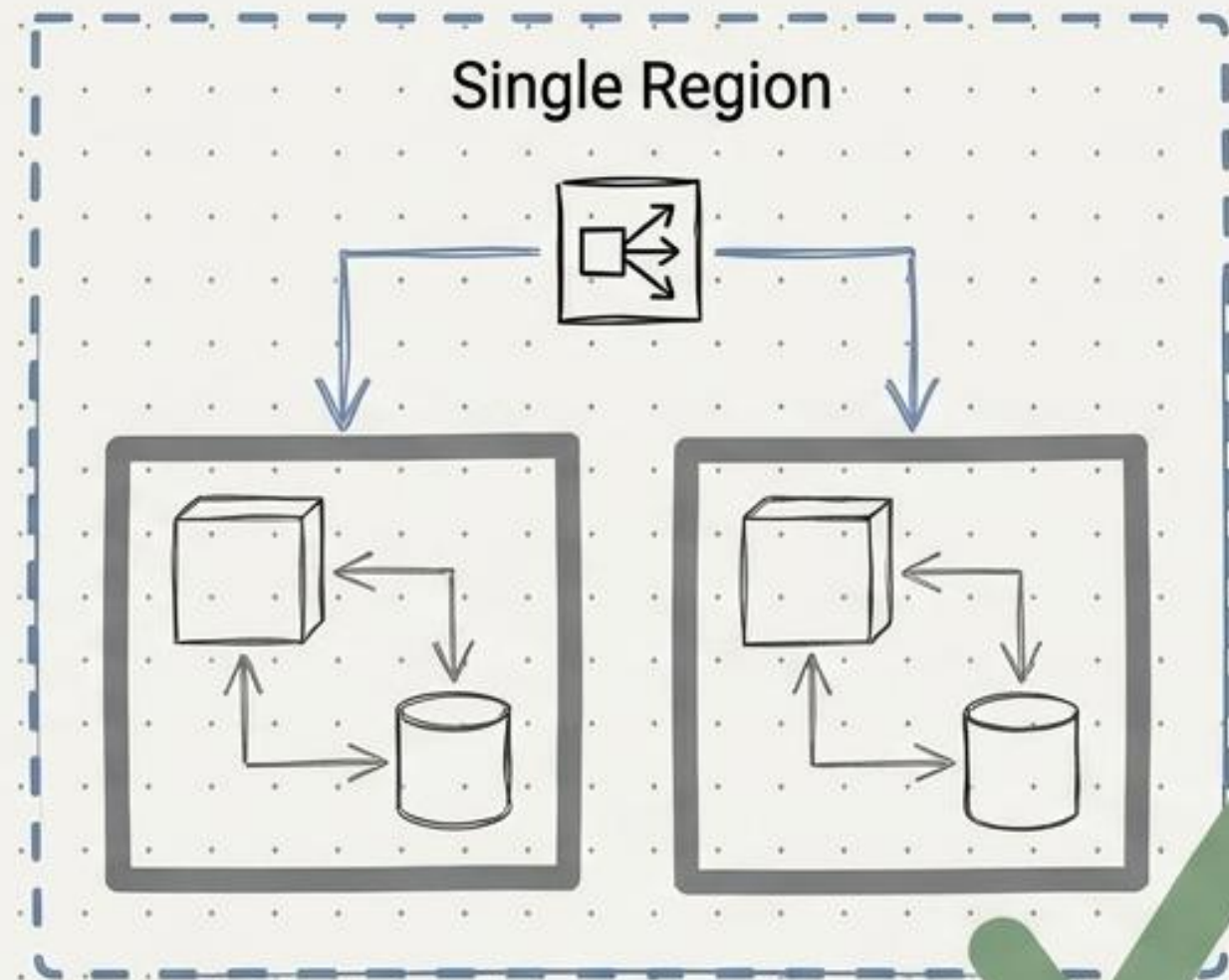


- Tighter objectives cost more, monotonically.
- An RTO that has never been proven in a failover drill is an aspiration, not an objective.

Gate 0: Do not build across regions on a fragile foundation



Anti-Pattern



Baseline



The **Prerequisite**: Multi-AZ within a single Region already protects against data center failure and is the AWS-recommended baseline for high availability.

Fix single-region resilience before jumping to multi-region.

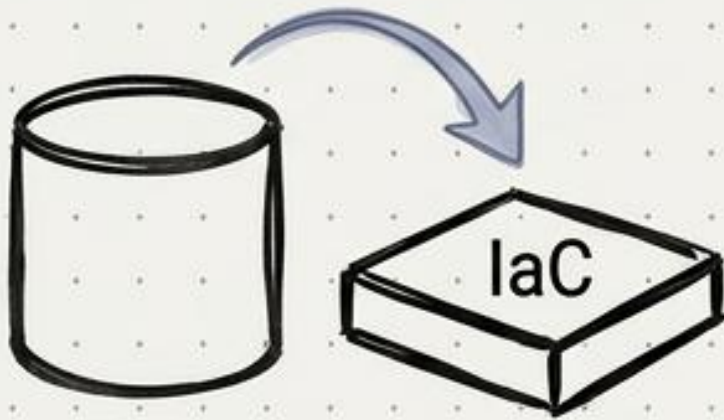
Gate 1: Classify the workload through Business Impact Analysis



T1 (Mission Critical)	Payments, patient safety	RT0: <15 min	RPO: Secs to near-zero
T2 (Business Critical)	Core APIs, fulfillment	RT0: 15m - 1h	RPO: Secs to mins
T3 (Important)	Internal tools, batch	RT0: 1-24h	RPO: Mins to hours
T4 (Deferrable)	Dev/test, sandboxes	RT0: >24h	RPO: Hours to 24h

The four target DR strategies on AWS

Backup & Restore (\$)



(Restore + rebuild from IaC)

Pilot Light (\$\$)



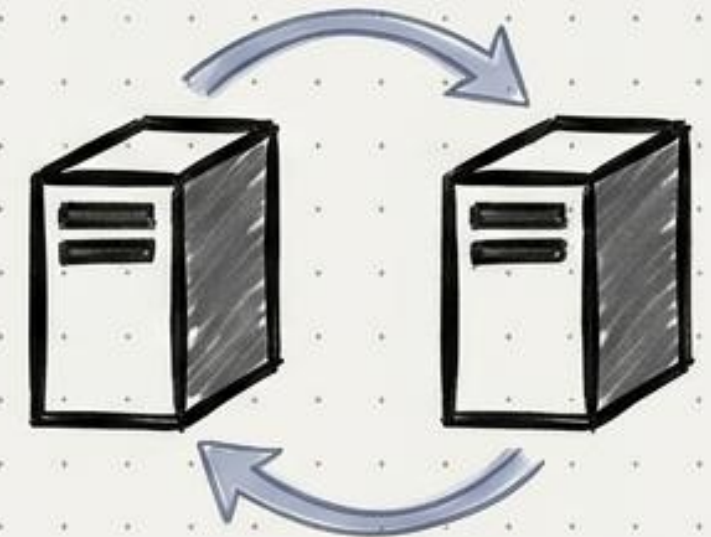
(Provision/scale, then shift traffic)

Warm Standby (\$\$\$)



(Scale up + shift traffic. Note: Can serve requests immediately at reduced capacity, unlike Pilot Light)

Multi-Site Active/Active (\$\$\$\$)



(Remove failed region from rotation)

Rough cost bands of production spend: B&R (~10-20%), Pilot Light (~20-40%), Warm Standby (~40-70%), Active/Active (~100-200%+).

Gate 2: The Core Decision Matrix (RTO Band × RPO Class)

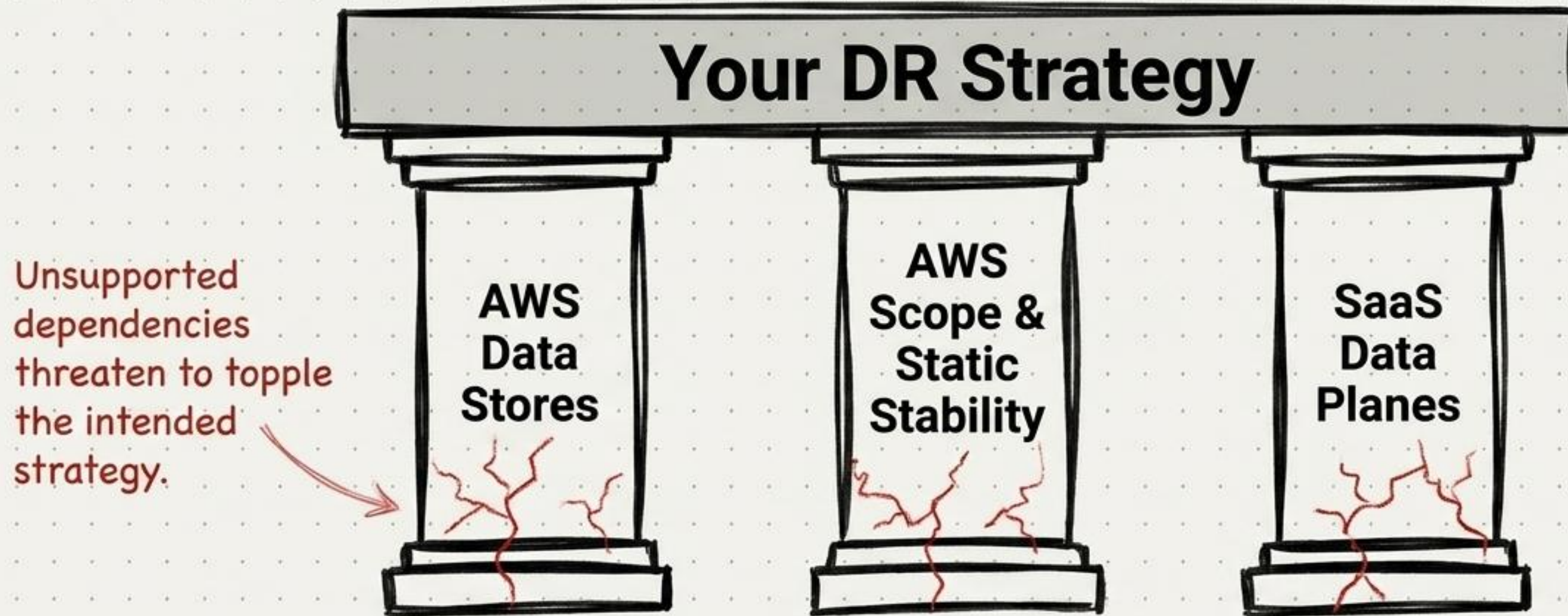
	POF	SOD	No Recovery
≤ 2h	Active/Active or Warm Standby		
2–4h	Pilot Light		laC Rebuild
4–24h	Pilot Light	Backup & Restore	laC Rebuild
24–48h	Pilot Light	Backup & Restore	laC Rebuild
48–72h	Pilot Light	Backup & Restore	laC Rebuild

Rule 1: RPO sets data posture floor and ceiling (e.g., SOD caps spend at daily backups; Aurora Global DB here is an over-engineering red flag).

Rule 2: RTO sets compute posture.

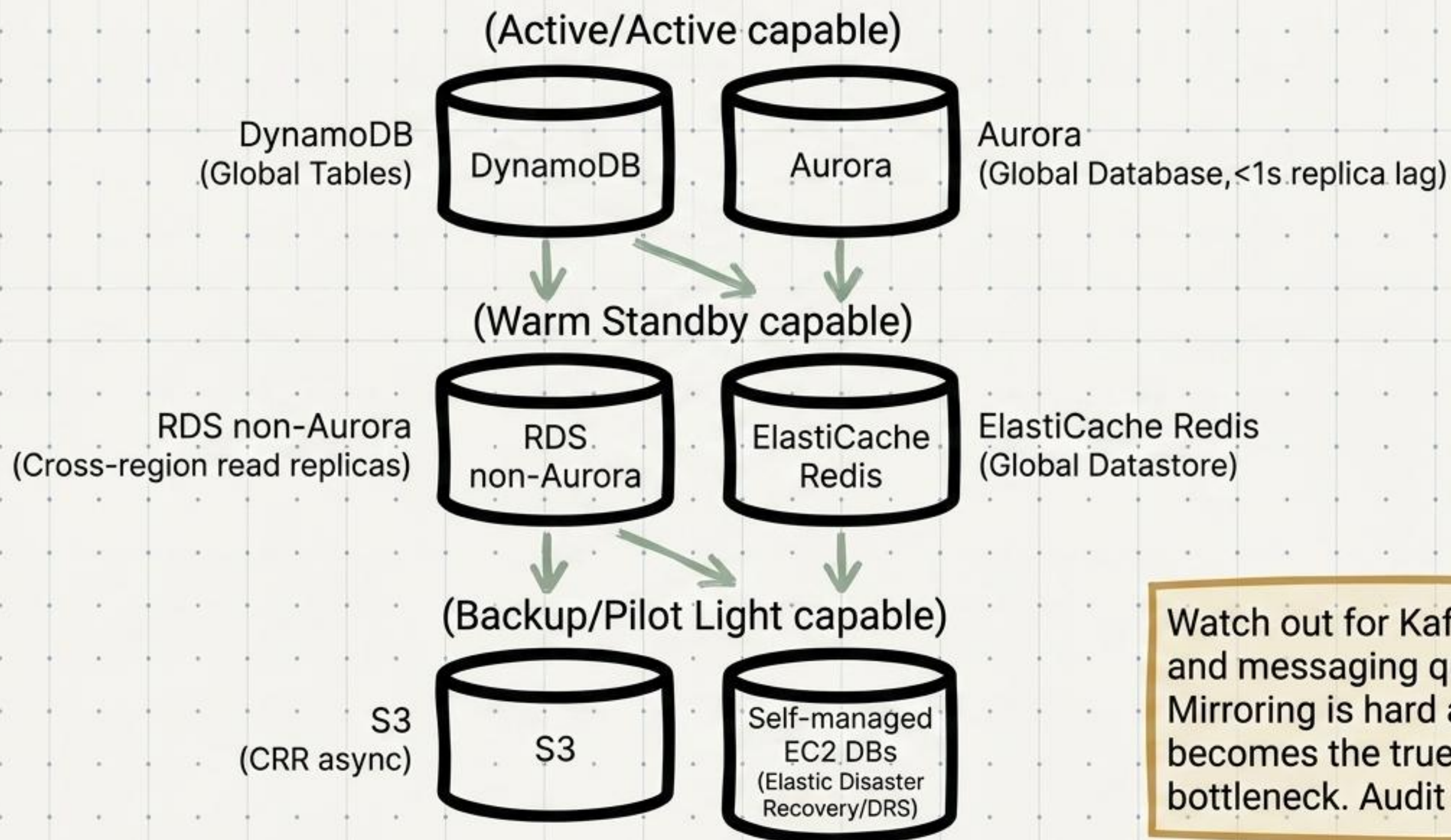
Gate 3: The Dependency Reality Check

Dependency Triad



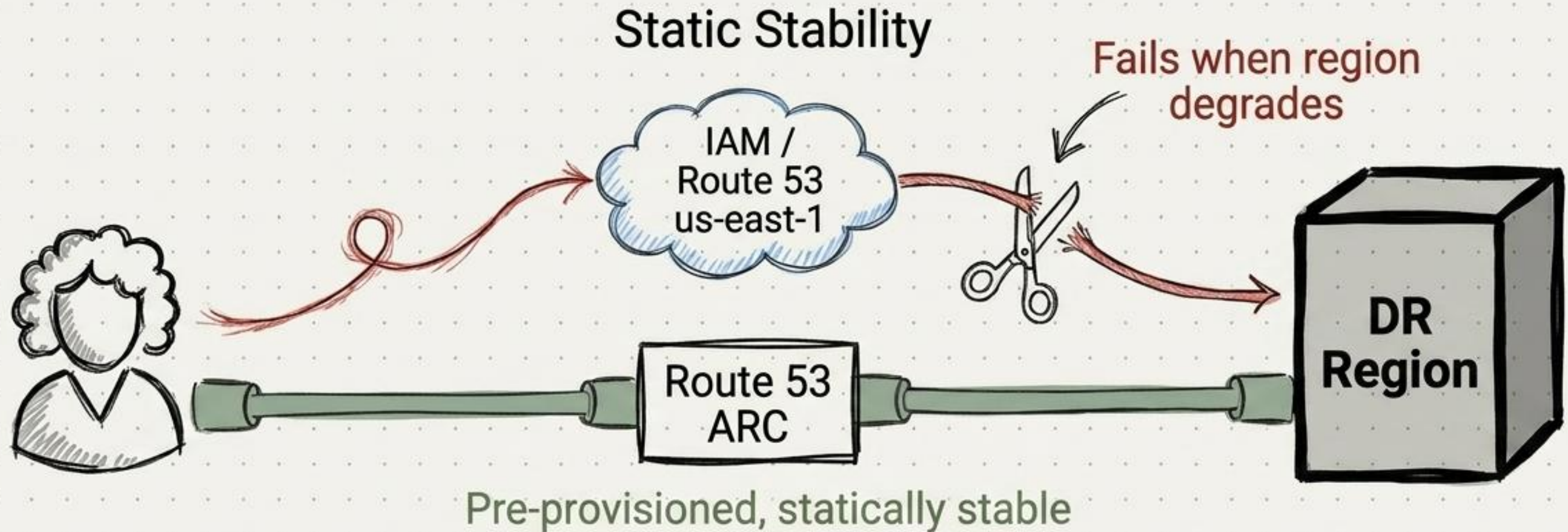
- Your strategy is capped not by intent, but by what your dependencies can do across regions.
- Effective objectives are computed, not declared: Effective RPO = worst across your data planes. Effective RTO = $\max(\text{own failover}, \text{slowest vendor failover})$.

Gate 3A: AWS Data Stores limit your ceiling



Watch out for Kafka/MSK and messaging queues: Mirroring is hard and often becomes the true RPO bottleneck. Audit carefully.

Gate 3B: Beware the Control Plane disconnect



Global Services

(IAM, Route 53) are typically homed in one region. Failover must depend on their globally distributed data planes only.

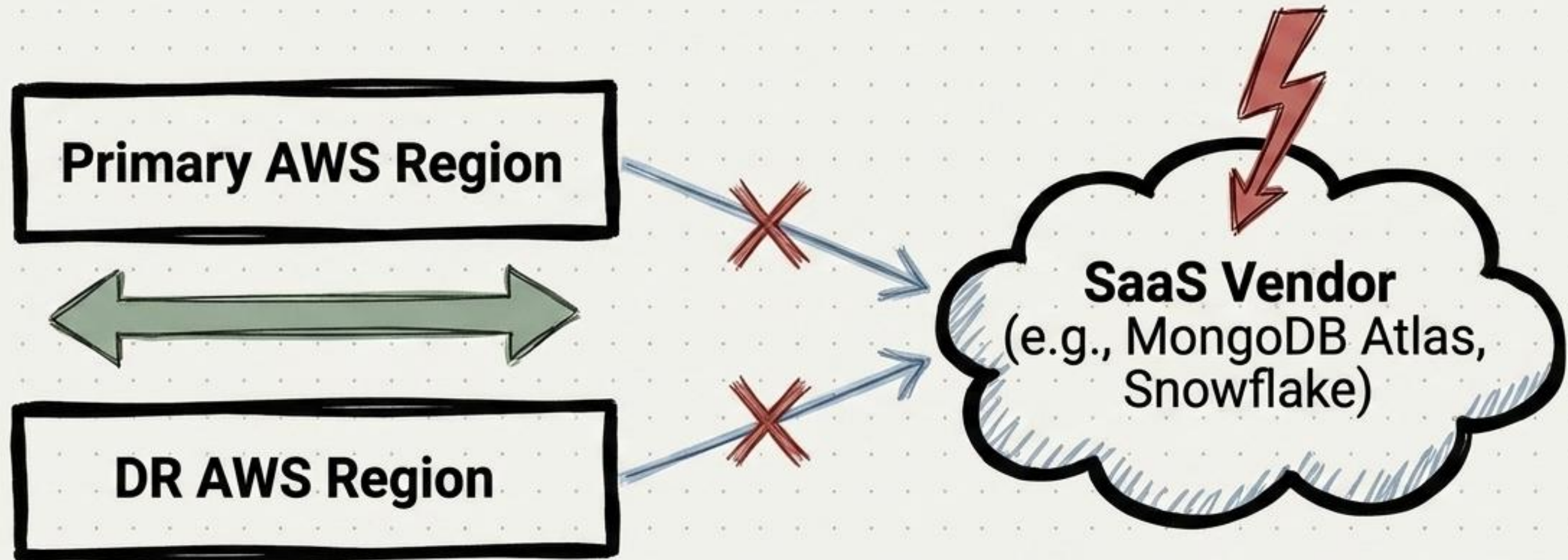
Regional Services

(EC2, KMS) need configured peers in the DR region.

Zonal Services

(EBS, Local Zones) are pinned to physical locations. Recovery requires snapshot/rebuild.

Gate 3C: The SaaS Correlated Failure Blindspot



SaaS products sit outside your AWS account but inside your RTO/RPO math.

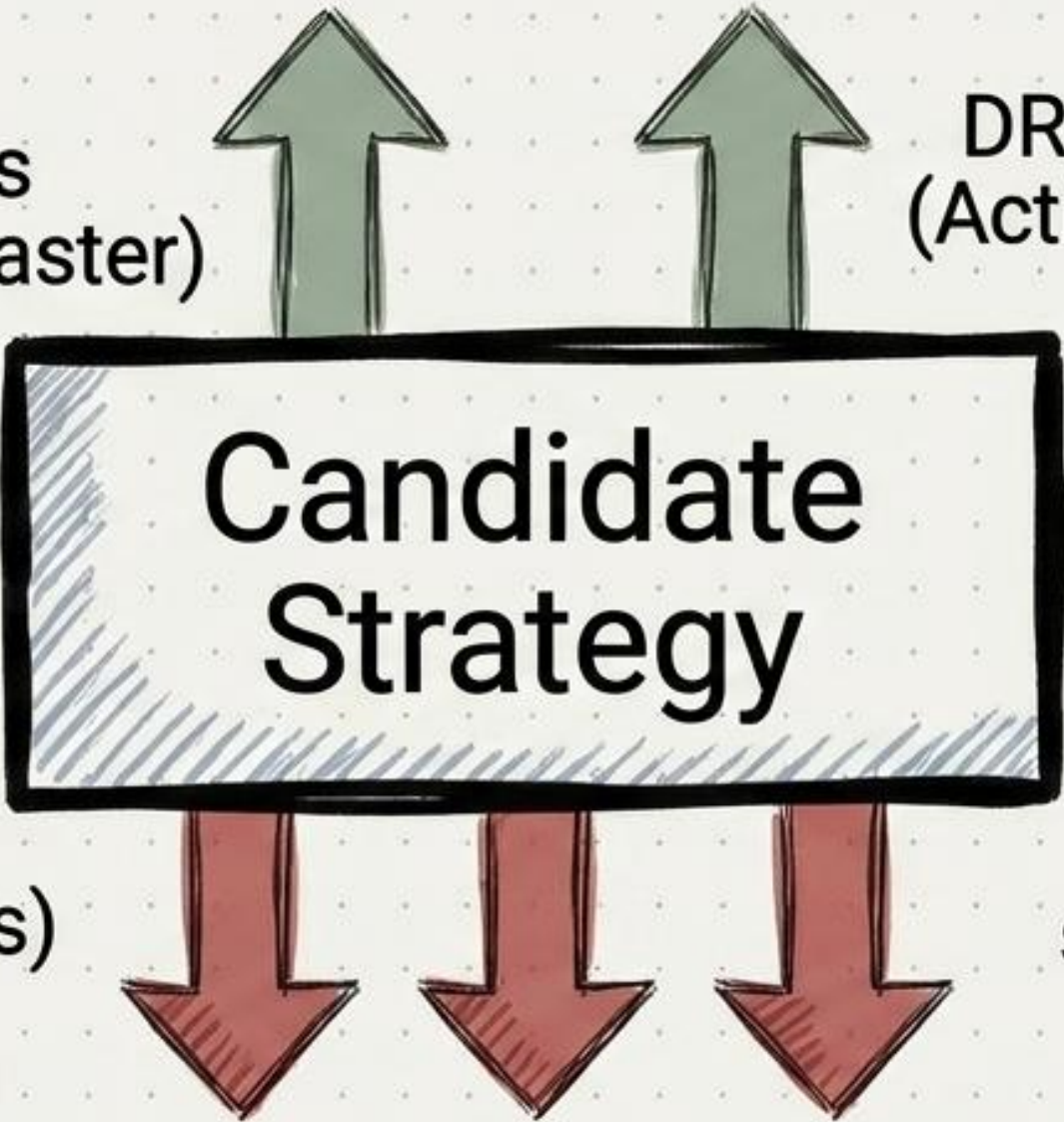
A vendor homed in your primary AWS region, without replication configured, makes your regional DR strategy fiction.

Are the network paths (PrivateLink) and Auth paths (SSO) pre-provisioned in the DR region?

Gate 4: Modifiers that alter your baseline

Everything in IaC
+ Automated Pipelines
(Pilot Light gets cheaper/faster)

DR budget > 100% of prod
(Active/Active becomes viable)



Candidate
Strategy

Zero Ops Maturity
(no runbooks, no game days)

Hard dependency on
single-region third party

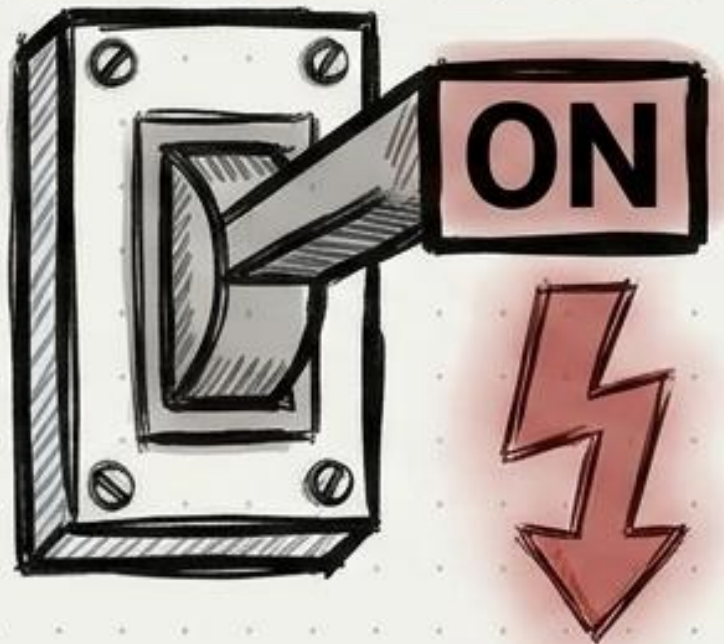
You cannot operate what
you cannot drill. Poor ops
maturity forcibly
downgrades your
achievable strategy.

Compliance / Data
Residency Limits

If static stability is
required, you must
pre-provision capacity.

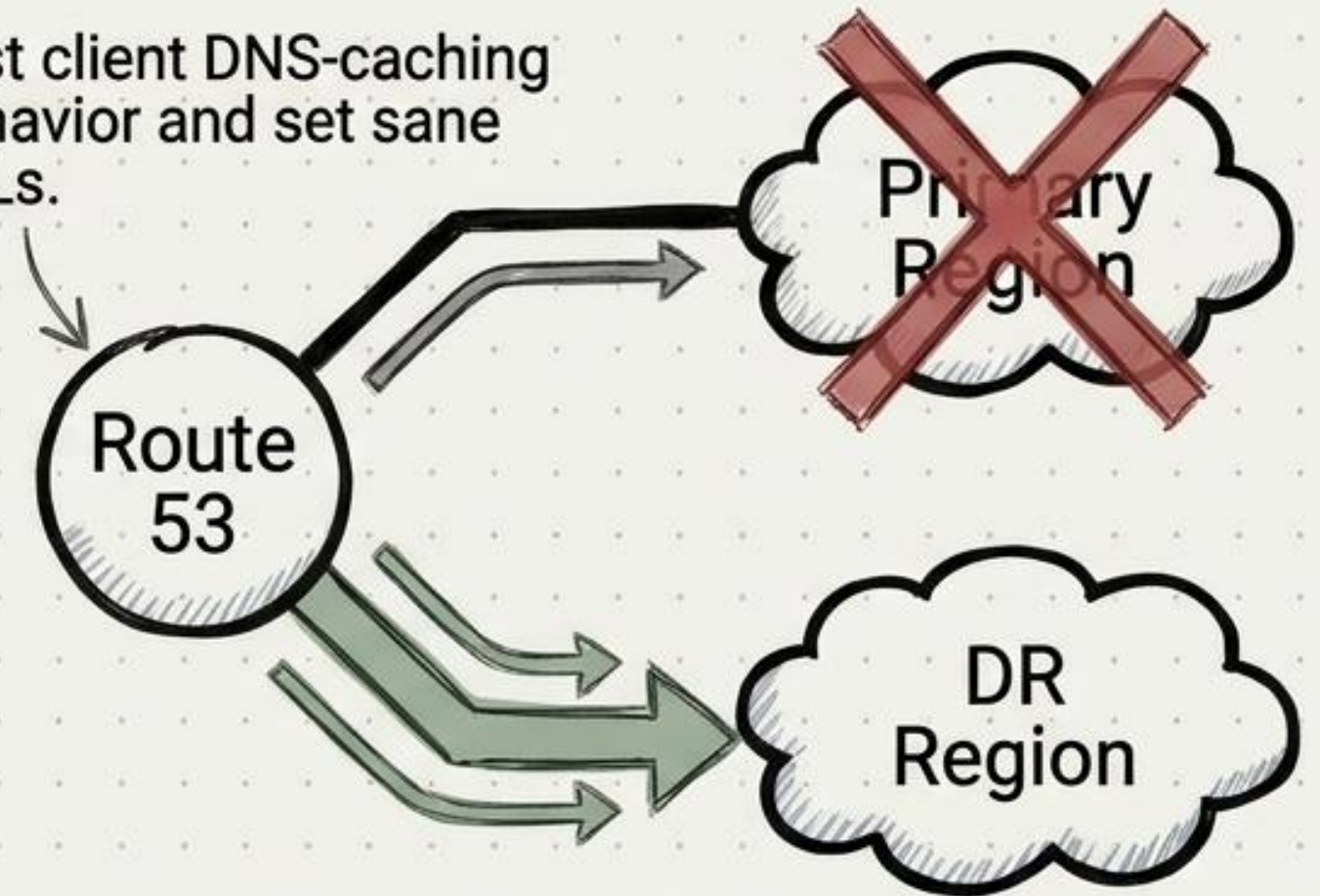
Gate 5: Failover Mechanics & Anti-Patterns

Decide who pulls the trigger.



Automated for Tier 1 (with anti-flap guardrails), Human-approved for lower tiers.

Test client DNS-caching behavior and set sane TTLs.



✗ Anti-Pattern: Launching new resources via control-plane APIs during a tight RTO. Pre-provision or pre-scale instead.

✗ Anti-Pattern: Replicating data corruption with no Point-in-Time Recovery (PITR).



Failback is a plan, not an afterthought. Data reconciliation after failover is usually harder than the failover itself.

Gate 6: Prove it, or you don't have it

The architecture is theoretical until proven in a game day.

Record capability facts once in a central DR Catalog; let the application teams consume, not research.

One-Page Worksheet

Tier/ RTO / RPO: _____

Budget Ceiling: _____

Gate 0 Result: _____

Worst Data-Layer RPO across dependencies: _____

Modifiers Applied: _____

Final Strategy: _____

Trigger Owner: _____

Last Measured Drill: _____

Cadence:
Tier 1: Quarterly game days.
Tier 2: Semi-annual.
Tier 3/4: Annual restore tests.